contorno

Contorno Indicator: the desktop application

How the hide-area method is implemented, and the tools that build and check it

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Abstract

The companion article measures the surface of an animal hide from one photograph. This paper describes the program that carries that method to an operator: **Contorno Indicator**, a desktop application on wxWidgets 3.2. The measuring code lives in five C++20 libraries that do not know the user interface. The application adds a window, a worker thread, seven languages, and an inspection file that keeps the photograph, the mask and the outline. OpenCV and LibTorch stay behind one library. A build without LibTorch still compiles and says, on screen, that it cannot measure. The program is configured with CMake and Ninja, through buildkit 2.0.0, and checked by GoogleTest under CTest. The release build of this tree registers 399 tests.

1. What this paper is

Two documents describe Contorno, and they answer different questions.

The article *Measuring the surface of animal hides from a single photograph* (`docs/articles/contorno-area-measurement/`) is the method: which pixels belong to the hide, and what area those pixels stand for. Its Python scripts implement that method again, for the figures and the uncertainty budget. They do not call the C++ libraries.

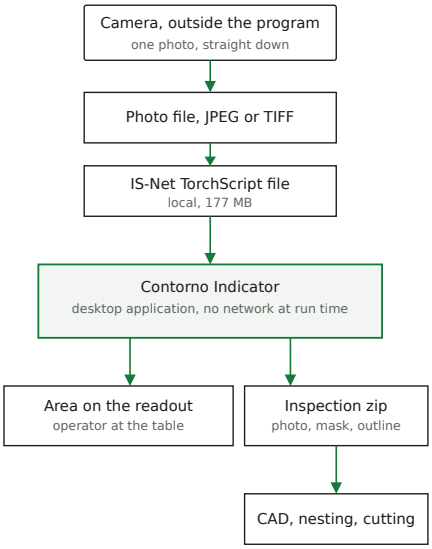
This paper is the program an operator runs. It states how the method was allocated to libraries and to the window, which interfaces the parts share, and which tools produce the executable and the tests. `docs/architecture/architecture.md` is the document for someone who is about to change the code. Where a count can drift, this paper gives the date it was read.

The statements of versions, paths and test counts were read from this tree and from the development machine on 7 November 2026. The physics (the scale formula, the bias of the outline, the cases where the background step fails) is not re-measured here. Those numbers stay in the article.

2. Purpose and boundary

Hides are traded by surface. Indicator shows that surface from a photograph taken straight down onto a table, and it keeps the photograph as the evidence of the measurement.

The program does not drive the camera, does not move the hide, and does not use the network while it runs. The IS-Net file and the manual are local files. Downstream CAD, nesting or cutting is outside the program: it receives the outline as DXF inside the inspection archive.



The program does not drive the camera
and does not move the hide.

What sits inside the program, and what stays outside it.

The operator at the table needs a number and a way to save the job. Whoever buys or sells the hide needs that number to be the area of the saved mask, not a second computation. The cutting side needs the outline in millimetres, in a file a CAD program can open.

3. Rules the implementation is built to

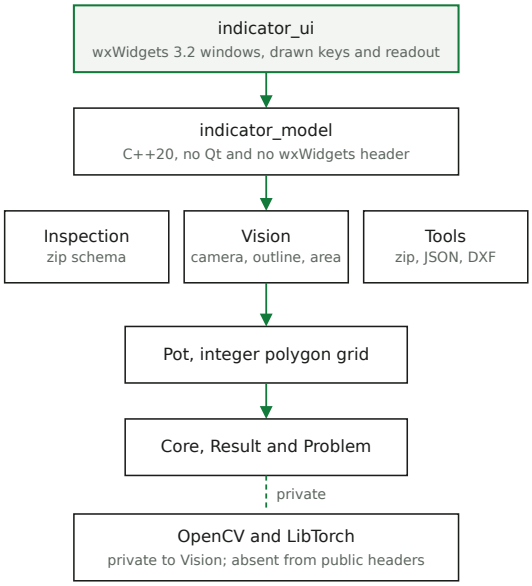
There is no separate requirements database. The rules below are the ones the build and the tests already enforce. Each names the check.

Rule	Where it is held
Libraries do not depend on a UI toolkit.	No Qt and no wxWidgets in any `libraries/*/CMakeLists.txt`.
OpenCV and LibTorch never appear in a public header.	Both are private dependencies of Vision. Only `Ai_impl.cpp` includes torch.
A build without LibTorch still compiles, runs and passes its tests.	`BUILD_VISION_AI`, `ai_available()`. Tests that need the model skip.
The application's model layer includes no widget header.	`indicator_model` links only the Contorno libraries. `WxLayers.ModelHasNoToolkit` reads the sources.
Library errors are a code plus values, not an English sentence.	`Core::Problem`. The application translates by code (`LibraryText`).
Public library functions do not throw across the boundary.	`noexcept` wrappers return `Result<T>`.
Without the model, the program measures nothing and says so.	The *Unavailable* state. No second, silent outline method.
The number on the readout is the area of the saved mask.	Area is `countNonZero(mask) × s_x × s_y`. The self-test compares the readout with that area.
The original photograph is stored untouched.	The inspection zip keeps `original.<ext>` byte for byte. No annotated image is written.

The last three are product rules. The first six are there so a change to the window cannot silently change the measurement, and so a machine without LibTorch is still a coherent build.

4. How the program is divided

```
Contorno/
  libraries/            measuring core, no UI toolkit
    Core/               Result<T>, Problem
    Pot/                integer polygon geometry
    Tools/              zip, JSON, DXF
    Vision/             camera, EXIF, segmentation, outline, area
    Inspection/         the inspection zip
  applications/
    Indicator/          the desktop application
      src/model/        thread, settings, messages
      src/ui/           wx windows
```



Dependencies, top-down. The arrow is “uses”.

Vision is the measuring core. Inspection owns the file format and calls Vision and Tools. The application is two static libraries and then the executable `Indicator`.

`indicator_model` is plain C++20. Text is UTF-8 `std::string`. Paths are `std::filesystem::path`. It links Tools, Vision and Inspection. The seven string tables (`strings/indicator_<lang>.cpp`) compile against this library, so a language can be tested without opening a window.

`indicator_ui` is the wxWidgets layer: the main window, the set-up dialog, the file dialog, the theme, and the controls that are drawn by hand. It is the only place that includes wx.

Anything that must be true for a saved inspection — the scale, the mask, the outline, the area — sits below the toolkit. Anything that is only true of a screen — colours, languages laid out left to right or right to left, which button is pressed — sits above it. The worker thread is on the model side and receives a function that posts its result back. The UI passes `wxTheApp->CallAfter`, so the result arrives on the thread that owns the widgets. Tests of the model pass a function of their own and never open a display.

5. Implementation methods

5.1 Error reporting

A public call returns `Result<T>`: either a value or a `Problem`. A problem carries a code (`camera.scale`), an English template used as a fallback, and named values (`{path}`, `{range}`). The application does not show the template. `LibraryText` maps the code into the current language. Paths and text that came from an exception stay as they arrived.

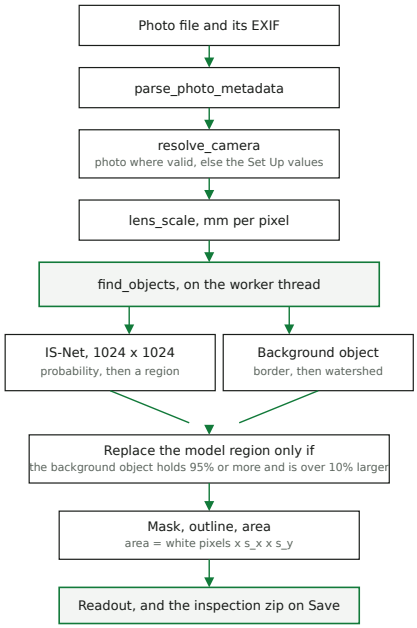
Wrong-side access of a `Result` (reading the value of an error, or the error of a value) aborts the program, and a test covers it. A forgotten check therefore stops the program instead of passing on a wrong value, and a library never has to know the operator's language.

5.2 Integer geometry

Pot stores polygons on a grid of 1 µm (`units_per_mm = 1000`): `Point64`, a closed ring, an outer ring counter-clockwise and holes clockwise. Integer coordinates make the saved outline bit-for-bit reproducible. The millimetre values an operator sees are a conversion at the boundary, not the representation the file is written from.

5.3 The chain from the file to the number

The physics is the article's. The program runs it as one sequence.



From the photograph to the number. The two outline paths meet at one rule.

Distance always comes from the Set Up dialog. Focal length is taken from the photograph when it falls in [0.1, 2000] mm, and the pixel size when it falls in [0.1, 100] µm; otherwise the setting is used and the camera line says why. The scale is

$$s_x = d \cdot p_x / f, \qquad s_y = d \cdot p_y / f.$$

For a photograph shown rotated by 90°, x and y are swapped. This is the thin-lens scale of a plane parallel to the sensor. It ignores focus travel and distortion. Both limits are stated in the article and in the manual; they are not corrected in the code.

The outline is IS-Net, exported once to TorchScript and run on CPU. `Ai_impl.cpp` converts the image to RGB, resizes it bilinearly to 1024 × 1024, applies (x/255 - 0.5), and runs the module. The module is cached per path behind a mutex. The output is checked

for shape and for finite values, normalised, and resized back. A missing file, a wrong shape or a non-finite output is a failed analysis, not a zero area.

The mask then follows the article's second step. A threshold of 0.5, a cleaning of one-pixel spurs, the largest region, and holes kept only when they are large enough and the model sees the table through them. On a spotted hide the network can take the dark patches for the object. A classical background step, on a shrunk and median-filtered copy, may replace that region when it holds at least 95 % of it and is more than 10 % larger. The window then adds a line asking the operator to check the outline. Where the hide runs along much of the frame, the rule usually keeps the model's region. A hide that only touches the frame in places can still have its region replaced.

The area is the count of white pixels in that mask, times (s_x) and (s_y). The outline written to DXF is that same mask, simplified within 0.5 mm, so the readout, the PNG and the DXF describe one region.

5.4 One analysis at a time

The GUI thread owns every widget. `Analyser` runs `find_objects` on one worker thread. A request made during a run waits. A newer request replaces the one that is waiting. The result of a run that a newer request has made stale is dropped, and the readout stays dimmed until the new result arrives. The worker does not touch widgets. It hands a task to the `Post` function it was given, and that function must run the task asynchronously: the task joins the worker that posted it.

The destructor drops a waiting request and joins the worker. A run already inside LibTorch is not cancelled. Shutdown waits for it, then discards the result. `WxWindow.CloseWhileAnalysingIsClean` covers that path.

At start the program loads settings, language and theme. The states that follow are the ones the window actually implements.

State	Readout	Save
No photo	dashes	refused
Analysing	previous value, dimmed	refused
Result	area	allowed
Saved inspection, reopened	the saved area	allowed
Failed	dashes, and the reason	refused
AI unavailable	dashes, and why	refused

Opening a saved inspection does not run the model again. The area on screen is the area in the file. A new photograph starts an analysis.

5.5 The inspection file

Save writes a zip, through Tools, with stored entries only. The parser checks the CRC, and it rejects absolute names, `..`, duplicates, encryption, split archives and archives over the size limit. The order is fixed:

Entry	What it is
`original.<ext>`	the photograph's bytes, untouched
`mask.png`	8-bit grey, object 255, same size as the photo
`camera.json`	distance, focal length, pixel size, and whether each came from the photo or from settings. Schema `contorno-camera`, version 4
`parameters.json`	minimum area and the 0.5 mm outline tolerance
`objects.dxf`	outer ring and holes, layer `OBJECT`, image millimetre frame
`result.json`	image size, the two scales, the area, the rings

The image millimetre frame has x to the right and y up, origin at the bottom-left of the photograph. A pixel column (c) and row (r) in an image of height (H) maps to

$$((c + 0.5)\, s_x\,,\ (H - r - 0.5)\, s_y\,).$$

Reading still accepts camera.json versions 1 to 3, as square pixels, and older hide inspections. Those older files yield the photograph and the camera only, and they are analysed again. `mask.png` must match the image size and contain only 0 and 255.

Settings are a separate, smaller file. On Linux, `WxConfigStore` asks wx for the user config directory. That directory follows `XDG_CONFIG_HOME`, and the file under it is `indicator.ini`, in `Contorno/Indicator`. The self-test does not use that file. It uses an in-memory store, so a test run cannot change the operator's camera.

5.6 The window

The main window is a photograph, a second linked view, a control panel and a status bar. The panel carries the wordmark, four keys, the mask toggle, the message log and a seven-segment readout. Those keys, the readout, the log and the progress bar are drawn by the application. Native controls are used where GTK will take the theme's colours. The file dialog is a `wxDialog` around `wxGenericFileCtrl`, because the GTK file chooser does not follow the theme.



The window after the sample cowhide, captured for the user manual on 6 November 2026. The yellow numbers are the manual's labels. The log's last line is the program's own timing of that analysis, 2.9 s. The hide is "Stretched Cowhide Skin" by Enda012, Wikimedia Commons, CC BY 4.0, reduced to 1600 × 1200.

The window icon is compiled into the binary. A path to a PNG is useless once the process is not started from the source tree, and under Wayland the desktop does not ask the window for a file. The taskbar icon comes from the desktop entry `contorno-indicator`, which is also the application id (`WM_CLASS` on X11, the Wayland app id).

Seven languages are compiled in: English, German, Spanish, Italian, French, Dutch and Arabic. Strings are keyed (`message.ai.no_model`) and carry `{placeholders}`. Numbers with units are not translated: Latin digits, a point as the decimal mark. Arabic sets right-to-left on the dialogs, the log and the key labels. The main window's geometry does not mirror. The log stores messages, not already rendered text, so a change of language rewrites old lines and keeps their times.



Set Up: sensor size or pixel size, distance, focal length, the scale those values give, and the language. Captured for the manual. The yellow numbers are the manual's labels.

The question mark in Set Up opens the manual in the dialog's language. The file is searched next to the executable, then in the source tree, then in English. A Chromium-family browser is started as an app window with a private profile, 1200 × 800, centred

on the dialog's screen. If none of those browsers is present, the system browser is used. If that fails, the warning names the file.

wx's own strings, those of the file dialog, are not in our tables. `WxCatalogue.cmake` compiles wx's catalogues into the executable for German, Spanish, Italian, French, Dutch and Arabic. It prefers the machine's `wxstd.mo`. If that is missing it downloads the `wx` 3.2.8 `.po`, checks a pinned SHA-256, and compiles it with `msgfmt`. Offline, or without `msgfmt`, those strings stay in English and the build warns.

5.7 What a failure does

The pattern used throughout is: detect a failure at the boundary that can see it, refuse the measurement or the file, and leave a message the operator can read in the current language.

Failure	What the operator gets
Built without LibTorch, or the model file is missing	The photograph, no area, and a line that says the outline is unavailable and where the file was sought.
Model corrupt, wrong shape, or non-finite output	The analysis fails. The previous result stays.
Unreadable photograph	The import fails. The previous state stays.
Camera value in the file outside the accepted range	The setting is used, and the camera line says the file value was rejected.
Damaged or hostile zip	Open is refused, with the entry and the reason.
A result that a newer import has superseded	Dropped. The readout stays dimmed.
Nothing on the table above the minimum area	Success, area 0, "No object found".
The model kept only part of a spotted hide	The background object is measured, and a line asks for the outline to be checked.
No browser for the manual	A warning that names the manual file.

The case that is deliberately not handled by a fallback is the missing model. A classical-only outline was tried in the work that led to the article, and it was not accurate enough where the hide is close in colour to the table. Shipping it as a silent spare would put a different instrument behind the same readout.

6. Tooling

The language of the program is C++20. The build is CMake (3.29 or newer; 3.31.6 on the development machine) with the Ninja generator (1.12.1 here). The compiler this tree is developed with is g++ 14.2.0 on Debian 13. Those three version numbers were read on the development machine. The versions the build pins are in the tables below.

buildkit 2.0.0 supplies the warnings, GoogleTest, the coverage flags, the install layout and the version taken from the git tag `vX.Y.Z` (fallback 1.0.0). `cmake/buildkit.cmake` uses `BUILDKIT_DIR`, else a checkout at `../buildkit`, else a fetch of tag `v2.0.0`. Contorno replaces buildkit's library and application macros so that a header-only library and the wx entry point fit the tree. The executable is created with `create_application(ENTRY WX_ui)`.

Presets live in `CMakePresets.json`. Each has its own directory under `build/`.

Preset	Build type	What it adds
`default`	Release	the ordinary build
`debug`	Debug	
`coverage`	Debug	`ENABLE_COVERAGE` (gcov / lcov)
`asan`	Debug	AddressSanitizer and UndefinedBehaviorSanitizer
`ci`	Release	the continuous-integration directory

wxWidgets is an installed dependency, found the way buildkit finds a development package. This machine has 3.2.8, which is also the version whose translations are pinned. On Linux the UI links GTK 3 (3.24.49 here) so that text fields take the theme's frame and selection colours. That link is Linux-only. The window tests set `GDK_BACKEND=x11`, because they read pixels back from the window.

OpenCV is the same kind of dependency: the installed library, or a fetched archive with a pinned URL and SHA-256 when `CONTORNO_OPENCV_FETCH` is on. This machine has OpenCV 4.10.0. OpenCV is used inside Vision for the image, the contours and the morphology. It is not a public type.

LibTorch is the CPU build 2.12.0. CMake looks in `/opt/libtorch` and `Torch_DIR`, then in the build tree, and otherwise downloads a zip whose SHA-256 is in `cmake/TorchSupport.cmake`. A failed download warns and continues. Vision then builds with `ai_available()` false. The IS-Net file is not part of `all`. The target `isnet_model` builds it in a Python virtual environment with `torch==2.12.0`, from the DIS weights, URL and hash pinned, using the article's export script. The C++ runtime and the exporter are the same PyTorch minor version on purpose. The PyTorch documentation marks TorchScript as deprecated in favour of `torch.export` (read on 7 November 2026). Loading it is confined to `ai_probability_impl`, which is the function a later runtime would have to replace.

Other tools, used at build or check time and not linked into the measurement:

Tool	Role
Python 3	model export; `build_manual.py --check`; `check_sources.py`
GNU gettext `msgfmt`	wx's own translations, when the `.mo` is not already installed
`rsvg-convert`	PNG sizes of the desktop icon from the SVG
`update-desktop-database`	registers the build-tree desktop entry under `~/.local/share`
Pandoc and Chromium	this paper and the measurement article, HTML then PDF
GoogleTest	the unit tests
CTest	runs them, including the manual check and the self-test

`BUILD_WX_UI` defaults on. Without wxWidgets, or with that option off, Indicator and its tests are skipped and the libraries still build. `BUILDKIT_OFFLINE` refuses downloads. A model or a catalogue that cannot be fetched then degrades as described above, rather than failing the configure step.

The commands used on this tree:

```
cmake --preset default
cmake --build --preset default --target isnet_model   # once
cmake --build --preset default
ctest --preset default
```

7. Speed, and the machine it runs on

At run time there is no Python. Indicator is one native process. It links the Contorno libraries, wxWidgets, GTK 3 on Linux, OpenCV and the LibTorch CPU libraries. Those torch libraries sit beside the executable and are found by an RPATH of `$ORIGIN`. The IS-Net file and the manual are local files. The process does not open a network connection to measure a hide. Python is used to export the model, to check the manual, and to produce the measurement article. It is not on the operator's path.

The article's scripts implement the same method again, so the figures do not depend on this program. Their times and the program's times are therefore two readings, not one benchmark repeated.

The article's timings were taken on the same development machine as the C++ timings below (its Section 3.2 and Section 3.5):

What was timed	Time
rembg IS-Net only, cowhide 1600 × 1200, one call after a warm-up	1.64 s
rembg IS-Net only, black calf 1334 × 1602, same conditions	1.48 s
TorchScript forward alone, 1024 × 1024, 4 threads, median of 5 after a warm-up	1.822 s
Whole Python analysis, network and background check, median at 1600 × 1200	2.3 s
The same, median at 4000 × 3000	7.5 s

The C++ path was timed on 7 November 2026 on the development machine, an Intel Core i7-10510U (4 cores, 8 threads, no GPU used). A small program linked to the release build of Vision called `find_objects` with the background check on, the outline tolerance at 0.5 mm, and a scale of 1 mm/px. The scale enters only the minimum-area filter, which turns a minimum in mm² into a pixel count; it does not change the network's input. One warm-up was discarded. LibTorch reported 4 threads, which is the default it chose on this CPU. The median of the five runs:

Photograph	Size	Median	The five runs
`cowhide_stretched.jpg`	1600 × 1200	2.69 s	2.56–2.81 s
`calf_black.jpg`	1334 × 1602	2.64 s	2.57–2.70 s

The manual's capture of that same cowhide, inside the real window, logged 2.9 s. One run, with the file read and the result posted back to the interface, beside a median of 2.69 s for the library call alone.

These timings are not one benchmark. The 2.3 s is a median over the article's set of 1600 × 1200 photographs, in Python. The 2.69 s is five runs of one file, in C++. The rembg lines are the network call only, so they sit closer to the 1.822 s forward than to a whole analysis. On a photograph of this size the native program takes about two and a half seconds, and most of that is the 1024 × 1024 forward.

8. How it is checked

Verification here means: does the program we built match the rules in section 3? Validation against a certified area, on real hides, is the open measurement of the article. It has not been done. The software tests do not stand in for it.

CTest on the release preset registers 399 tests, counted from the discovery files of `build/default` on 7 November 2026. They add as follows.

Suite	Tests	What it is responsible for
`test_contorno_core`	18	`Result`, `Problem`
`test_contorno_pot`	2	the 1 µm grid
`test_contorno_tools`	28	zip, including archives written by Python; JSON; DXF
`test_contorno_vision`	74	the scale, EXIF, the outline on the sample photographs, holes, spotted and speckled hides
`test_contorno_inspection`	14	the zip schema, older versions, legacy files
`test_indicator_model`	90	analysis, settings, languages, messages, the manual locator. No display
`test_indicator`	170	windows on a display: panel, readout, log, file dialog, set-up, languages, right-to-left
`indicator_self_test`	1	the real `Indicator` process
`indicator_manual`	1	every manual page, its links, and every label it quotes
`samples_sources`	1	every sample file has a row in `docs/samples/SOURCES.md`

The self-test is the integration check. It launches `Indicator --self-test` on `docs/samples/leather/calf_black.jpg` with `GDK_BACKEND=x11` and a timeout of 300 s. It imports the photograph and waits, checks that the readout shows the area of the result, imports again while an analysis is running and checks that the readout never shows a stale number, writes the zip and checks the six entry names and the DXF rings, opens that zip and checks that no new analysis starts and that the rings match, then writes a screenshot. Exit code 77 means the outline is unavailable (no LibTorch, or no model file). CTest treats 77 as a skip, not as a failure. The self-test uses the in-memory settings store.

The manual check is the other kind of integration. `build_manual.py --check` builds every page in every language and compares the labels it quotes with the string tables. A renamed key fails this test, which is the reason the manual and the translations can be reviewed as text.

What this suite does not do: it does not compare an area with a reference instrument, it does not run on Windows or macOS in this project's continuous practice (the desktop entry and the icon path are written for Linux; the executable has resource branches for Windows and macOS), and it does not treat the duration as a pass or fail. Section 7 is a

reading of that duration, not a limit in the test suite. The application still logs its own seconds with each result.

9. What is installed

`cmake --install` on Linux lays the program out as follows. Libraries are shared, and the LibTorch shared objects are installed beside them with an RPATH of `$ORIGIN`, so the executable finds them without a system-wide torch.

```
bin/Indicator
lib/libcontorno_*.so    and the LibTorch objects
share/contorno/models/isnet-general-use.pt
share/contorno/manual/
share/contorno/articles/contorno-area-measurement/
share/applications/contorno-indicator.desktop
share/icons/hicolor/.../contorno-indicator.png
```

The model, the manual and the measurement article are installed with the application. A developer build finds the model and the manual next to the executable first, then in the source tree, and does not need an install to run. Building `Indicator` also copies the desktop entry into the current user's `~/.local/share`, with `Exec` rewritten to the build-tree binary, so a Wayland session shows the icon before anything is installed.

10. Limits that belong to the system, not to one file

The measurements cited here were made in the article, not repeated for this paper.

The scale (d p / f) overestimates length by about (f/(d-f)): near +0.87 % in length and +1.76 % in area at 13 mm focal length and 1500 mm distance. A calibration on a known object removes it. The article also measured an outward bias of the outline, on the order of 0.6 % to 1.8 % of area on the synthetic photographs. A small calibration card inherits that bias in the wrong direction for a large hide; the manual asks for a board of about a square metre.

There is no distortion correction. The hide should stay central. Only the largest object is measured: two hides that touch become one area, which is a choice made so that one photograph gives one number. The background step fails when the border of the frame is not background. A mat smaller than the frame is the case the article shows, at about +80 % on one synthetic photograph.

TorchScript as a file format is the migration risk. The rest of the program never includes a torch header, so the risk is local, but it is real: a future PyTorch may refuse the file the exporter writes today.

The tests fix the behaviour the code has, including that of the background rule. A test passes as long as the rule behaves as coded, even on a hide that fills the frame, where the rule is wrong.

11. Rebuilding this paper

From `docs/articles/contorno-indicator/`:

```
scripts/build_paper.sh
```

The script runs Pandoc, then headless Chromium, and writes `contorno-indicator.html` and `contorno-indicator.pdf` beside the Markdown. It uses the stylesheet and the wordmark of the measurement article, so the two papers print as one series. The figures are the SVG files in `figures/`. They are drawn to be read down a narrow A4 column.

Statement on the use of generative AI

The author used an AI assistant (Claude, Anthropic) to draft and revise parts of the text. The author reviewed and revised the text and takes full responsibility for the content.